
Calculus & Real Analysis

Comprehensive Study Plan for BPS-17 Lecturer (Mathematics)

AJK Public Service Commission (AJKPSC)

Applied Mathematics Series

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✉ Applied Mathematics | BPS-17 AJKPSC Prep

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About This Study Plan

This document provides a **complete, exam-focused** study plan for the **BPS-17 Lecturer Mathematics AJKPSC/FPSC/PPSC/KPSC** paper. It covers:

- Real Number System
- Sequences & Series
- Limits & Continuity
- Differentiability
- Mean Value Theorems
- Riemann Integration
- Improper Integrals
- Functions of Several Variables
- Uniform Convergence
- Power Series & Taylor's Theorem

Each topic includes: **core concept** → **key formulas** → **MCQs with detailed explanations**.

 Suggested 8-Week Study Plan

Week	Topic	Focus
1	Real Number System & Sequences	ε - δ proofs, Completeness
2	Series of Real Numbers	Convergence tests, Power series
3	Limits & Continuity	IVT, EVT, Uniform continuity
4	Differentiability	MVT, Taylor's theorem
5	Riemann Integration	FTC, Improper integrals
6	Functions of Several Variables	Partial derivatives, Jacobian
7	Uniform Convergence	Weierstrass M-test, term-by-term
8	Full Revision + MCQ Practice	All past papers

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1 Real Number System

Core Concepts

Ordered Field

$\mathbb{R}$ is a complete ordered field. $\mathbb{Q}$ is ordered but *not* complete.

Completeness

Every non-empty subset of $\mathbb{R}$ bounded above has a **least upper bound** (supremum). This is the **Completeness Axiom** — what separates $\mathbb{R}$ from $\mathbb{Q}$.

Archimedean Property

For every $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $n > x$.

Density of $\mathbb{Q}$

Between any two real numbers there exists a rational number.

Intervals

(a, b) , $[a, b]$, $(-\infty, b]$, etc. A closed bounded interval is *compact*.

Key Results

$$\sup(A + B) = \sup A + \sup B \quad (1)$$

$$\inf(A + B) = \inf A + \inf B \quad (2)$$

$$\sup(-A) = -\inf(A) \quad (3)$$

$$\text{Cauchy-Schwarz: } \left| \sum a_k b_k \right|^2 \leq \left(\sum a_k^2 \right) \left(\sum b_k^2 \right) \quad (4)$$

PSC Exam Hotspots — Real Numbers

- Difference between **sup/inf** and **max/min** (sup may not be attained)
- $\sqrt{2} \notin \mathbb{Q}$ — classic proof by contradiction
- **Nested Interval Property**: if $[a_n, b_n] \supseteq [a_{n+1}, b_{n+1}]$ and $b_n - a_n \rightarrow 0$, the intersection is a single point
- $\mathbb{Q}$ is countable; $\mathbb{R}$ is uncountable (Cantor diagonal argument)

1.1 MCQs: Real Number System

Q1. The least upper bound of the set $A = \left\{ \frac{n}{n+1} : n \in \mathbb{N} \right\}$ is:

- (A) $\frac{1}{2}$ (B) 1 (C) 2 (D) Does not exist

✓ Answer & Explanation

Correct Answer: (B) 1

The elements are $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots$ which are strictly increasing and approach 1 as $n \rightarrow \infty$.

Since $\frac{n}{n+1} < 1$ for all n , the number 1 is an upper bound. For any $\varepsilon > 0$, choose

$n > \frac{1}{\varepsilon} - 1$; then $\frac{n}{n+1} > 1 - \varepsilon$, so no number less than 1 can be an upper bound.

Thus $\sup A = 1$.

Note: $1 \notin A$, so $\max A$ does not exist, but $\sup A = 1$.

Q2. Which of the following is **NOT** a property of $\mathbb{R}$?

(A) Every Cauchy sequence converges (B) $\mathbb{Q}$ is dense in $\mathbb{R}$ (C) Every bounded sequence has a convergent subsequence (D) Every non-empty bounded subset has a maximum element

✓ Answer & Explanation

Correct Answer: (D)

(A) is the **Cauchy completeness** of $\mathbb{R}$ — TRUE.

(B) **Density of rationals** — TRUE.

(C) **Bolzano-Weierstrass Theorem** — TRUE.

(D) FALSE. The set $\left\{ \frac{n}{n+1} \right\}$ is bounded above by 1, but its supremum (1) is not in the set, so it has no maximum. Having a *supremum* is guaranteed; having a *maximum* is not.

Q3. The set of irrational numbers is:

(A) Countable (B) Finite (C) Uncountable (D) Countably infinite

✓ Answer & Explanation

Correct Answer: (C) Uncountable

$\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$. Since $\mathbb{R}$ is uncountable and $\mathbb{Q}$ is countable, the irrationals $\mathbb{R} \setminus \mathbb{Q}$ must be uncountable. (A countable union of countable sets is countable; if irrationals were countable, $\mathbb{R}$ would be countable — contradiction.)

💡 PSC Exam Trick

PSC Favourite: “Which set is countable?” — Know: $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are countable; $\mathbb{R}, (0, 1)$, irrational numbers are uncountable.

2 Sequences of Real Numbers

Core Concepts

Convergence

$\{a_n\} \rightarrow L$ iff $\forall \varepsilon > 0, \exists N$ such that $n > N \Rightarrow |a_n - L| < \varepsilon$.

Cauchy Sequence

$\{a_n\}$ is Cauchy iff $\forall \varepsilon > 0, \exists N$ such that $m, n > N \Rightarrow |a_m - a_n| < \varepsilon$.

Monotone Convergence

Every bounded monotone sequence converges.

Bolzano-Weierstrass

Every bounded sequence has a convergent subsequence.

Squeeze Theorem

If $a_n \leq b_n \leq c_n$ and $a_n, c_n \rightarrow L$, then $b_n \rightarrow L$.

Limit Superior/Inferior

$\limsup a_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} a_k$; similarly for $\liminf$.

Standard Limits — Memorise These!

$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0 \quad (p > 0) \quad (5)$$

$$\lim_{n \rightarrow \infty} n^{1/n} = 1 \quad (6)$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x \quad (7)$$

$$\lim_{n \rightarrow \infty} \frac{n!}{n^n} \cdot e^n \cdot n^{-1/2} = \sqrt{2\pi} \quad (\text{Stirling}) \quad (8)$$

$$\lim_{n \rightarrow \infty} r^n = \begin{cases} 0 & |r| < 1 \\ 1 & r = 1 \\ \text{diverges} & |r| > 1 \end{cases} \quad (9)$$

2.1 MCQs: Sequences

Q4. The sequence $a_n = \frac{(-1)^n}{n}$ is:

(A) Convergent to 0 (B) Divergent (C) Bounded but not convergent (D) Monotone decreasing

✓ **Answer & Explanation**

Correct Answer: (A) Convergent to 0

$|a_n - 0| = \frac{1}{n} \rightarrow 0$. By the squeeze theorem (since $-\frac{1}{n} \leq a_n \leq \frac{1}{n}$ and both sides $\rightarrow 0$), the sequence converges to 0.

(D) is false — the sequence alternates sign, so it is not monotone.

Q5. $\lim_{n \rightarrow \infty} \left(1 + \frac{3}{n}\right)^n$ equals:

(A) e (B) e^3 (C) $3e$ (D) 1

✓ **Answer & Explanation**

Correct Answer: (B) e^3

Using the standard limit $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$ with $x = 3$:

$$\lim_{n \rightarrow \infty} \left(1 + \frac{3}{n}\right)^n = e^3$$

Q6. A sequence that is Cauchy in $\mathbb{R}$ is necessarily:

(A) Monotone (B) Bounded only (C) Convergent (D) Divergent

✓ **Answer & Explanation**

Correct Answer: (C) Convergent

$\mathbb{R}$ is **complete**: a sequence of reals is Cauchy if and only if it converges. Every Cauchy sequence is bounded, but in $\mathbb{R}$, bounded alone does not guarantee convergence (e.g. $(-1)^n$ is bounded but diverges). However, being Cauchy is the exact condition for convergence in $\mathbb{R}$.

Q7. Which of the following sequences **diverges**?

(A) $\left\{\frac{n^2}{n^2 + 1}\right\}$ (B) $\left\{\frac{\sin n}{n}\right\}$ (C) $\{(-1)^n\}$ (D) $\left\{\frac{1}{2^n}\right\}$

✓ **Answer & Explanation**

Correct Answer: (C) $\{(-1)^n\}$

(A) $\rightarrow 1$; (B) $\rightarrow 0$ (squeeze, since $|\frac{\sin n}{n}| \leq \frac{1}{n}$); (D) $\rightarrow 0$.

(C) oscillates between +1 and -1 with no single limit — diverges.

💡 PSC Exam Trick

For $\lim_{n \rightarrow \infty} a_n^{1/n}$ or ratio tests in sequences, relate to the root/ratio test for series. These appear in the same PSC question set.

3 Series of Real Numbers

📖 Convergence Tests — PSC Exam Essentials

Comparison Test

$0 \leq a_n \leq b_n$: if $\sum b_n$ converges, so does $\sum a_n$.

Ratio Test $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$: converges if $L < 1$, diverges if $L > 1$, inconclusive if $L = 1$.

Root Test $L = \lim_{n \rightarrow \infty} |a_n|^{1/n}$: same conclusion as ratio test.

Integral Test

$\sum_{n=1}^{\infty} f(n)$ and $\int_1^{\infty} f(x) dx$ converge or diverge together (for f positive, decreasing).

Alternating Series

$\sum (-1)^n b_n$ converges if $b_n \geq b_{n+1}$ and $b_n \rightarrow 0$.

Absolute Convergence

$\sum |a_n|$ converges $\Rightarrow \sum a_n$ converges.

p-Series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges iff $p > 1$.

📊 Standard Series Results

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \quad |x| < 1 \quad (\text{geometric}) \quad (10)$$

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty \quad (\text{harmonic, diverges}) \quad (11)$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \quad (\text{Basel problem}) \quad (12)$$

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x \quad (\text{exponential series}) \quad (13)$$

3.1 MCQs: Series

Q8. The series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges to:

- (A) 2 (B) 1 (C) $\frac{\pi^2}{6}$ (D) Diverges

✓ Answer & Explanation

Correct Answer: (B) 1

Use partial fractions: $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$.

This is a **telescoping series**:

$$S_N = \sum_{n=1}^N \left(\frac{1}{n} - \frac{1}{n+1} \right) = 1 - \frac{1}{N+1} \xrightarrow{N \rightarrow \infty} 1$$

Q9. By the ratio test, $\sum_{n=1}^{\infty} \frac{n!}{n^n}$:

- (A) Converges (B) Diverges (C) Test is inconclusive (D) Converges to e

✓ Answer & Explanation

Correct Answer: (A) Converges

Let $a_n = \frac{n!}{n^n}$. Then:

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{n^n}{(n+1)^n} = \left(\frac{n}{n+1} \right)^n = \left(1 - \frac{1}{n+1} \right)^n \rightarrow \frac{1}{e} < 1$$

Since $L = 1/e < 1$, the series **converges** by the ratio test.

Q10. The p -series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges for:

- (A) $p \geq 1$ (B) $p > 1$ (C) $0 < p \leq 1$ (D) All $p > 0$

✓ Answer & Explanation

Correct Answer: (B) $p > 1$

By the integral test: $\int_1^{\infty} \frac{1}{x^p} dx$ converges iff $p > 1$.

- $p = 1$: harmonic series, diverges.

- $p = 2$: converges to $\frac{\pi^2}{6}$ (Basel problem).
- $p = 1/2$: diverges.

4 Limits and Continuity



Core Concepts

ε - δ Limit $\lim_{x \rightarrow a} f(x) = L$ iff $\forall \varepsilon > 0, \exists \delta > 0$ such that $0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

Continuity f is continuous at a iff $\lim_{x \rightarrow a} f(x) = f(a)$.

IVT If f is continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$, then $\exists c \in (a, b)$ with $f(c) = 0$.

EVT A continuous function on a closed bounded interval $[a, b]$ attains its max and min.

Uniform Continuity

f is uniformly continuous on S if $\forall \varepsilon > 0$, one $\delta > 0$ works for *all* $x \in S$.



Standard Limits — Must Know

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad (14)$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2} \quad (15)$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \quad (16)$$

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1 \quad (17)$$

$$\lim_{x \rightarrow 0} (1+x)^{1/x} = e \quad (18)$$

4.1 MCQs: Limits and Continuity

Q11. $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$ equals:

- (A) $\frac{3}{5}$ (B) $\frac{5}{3}$ (C) 1 (D) 0

✓ Answer & Explanation

Correct Answer: (A) $\frac{3}{5}$

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{5x} = \frac{3}{5} \cdot \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} = \frac{3}{5} \cdot 1 = \frac{3}{5}$$

General rule: $\lim_{x \rightarrow 0} \frac{\sin ax}{bx} = \frac{a}{b}$.

Q12. $f(x) = \frac{|x|}{x}$ at $x = 0$ is:

(A) Continuous and differentiable (B) Continuous but not differentiable (C) Discontinuous (D) Undefined only

✓ Answer & Explanation

Correct Answer: (C) Discontinuous

$$f(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}; f(0) \text{ is undefined.}$$

Left-hand limit $= -1 \neq 1 =$ right-hand limit, so the limit does not exist and f is **discontinuous** at $x = 0$.

Q13. Every continuous function on $[a, b]$ is:

(A) Differentiable (B) Uniformly continuous (C) Monotone (D) Bounded but not necessarily uniformly continuous

✓ Answer & Explanation

Correct Answer: (B) Uniformly continuous

This is **Heine-Cantor Theorem**: A continuous function on a compact (closed and bounded) interval $[a, b]$ is uniformly continuous.

Counter-examples for the others: $f(x) = |x|$ is continuous but not differentiable at 0; $f(x) = \sin x$ on $[0, \pi]$ is not monotone.

Q14. By the Intermediate Value Theorem, $f(x) = x^3 - x - 1$ has a root in:

(A) $(-2, -1)$ (B) $(1, 2)$ (C) $(0, 1)$ (D) $(-1, 0)$

✓ Answer & Explanation

Correct Answer: (B) $(1, 2)$

$$f(1) = 1 - 1 - 1 = -1 < 0 \text{ and } f(2) = 8 - 2 - 1 = 5 > 0.$$

Since f is continuous and $f(1) < 0 < f(2)$, by IVT there exists $c \in (1, 2)$ with

$$f(c) = 0.$$

5 Differentiability



Core Concepts

Derivative $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ (if limit exists, f is differentiable at a).

Differentiable $\Rightarrow$ Continuous

The converse is false ($|x|$ at 0).

Rolle's Theorem

f cts on $[a, b]$, diff on (a, b) , $f(a) = f(b) \Rightarrow \exists c$ with $f'(c) = 0$.

MVT (Lagrange)

f cts on $[a, b]$, diff on $(a, b) \Rightarrow \exists c \in (a, b)$ with $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Cauchy MVT

$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$ for some $c \in (a, b)$.

L'Hôpital's Rule

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ when limit is $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

Taylor's Theorem

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k + R_n(x).$$



Taylor Series — Memorise

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \dots \quad (19)$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots \quad (20)$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \quad (21)$$

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots, \quad |x| \leq 1, x \neq -1 \quad (22)$$

5.1 MCQs: Differentiability & MVT

Q15. $f(x) = x^{2/3}$ at $x = 0$ is:

- (A) Differentiable with $f'(0) = 0$ (B) Differentiable with $f'(0) = 2/3$ (C) Continuous but not differentiable (D) Discontinuous

✓ Answer & Explanation

Correct Answer: (C) Continuous but not differentiable

f is continuous at 0 since $\lim_{x \rightarrow 0} x^{2/3} = 0 = f(0)$.

For differentiability: $\lim_{x \rightarrow 0} \frac{x^{2/3} - 0}{x - 0} = \lim_{x \rightarrow 0} x^{-1/3} = \pm\infty$. The derivative does not exist (vertical tangent).

Q16. By Rolle's Theorem, if $f(x) = x^2 - 4x + 3$ on $[1, 3]$, then $f'(c) = 0$ at:

- (A) $c = 1$ (B) $c = 2$ (C) $c = 3$ (D) $c = 4$

✓ Answer & Explanation

Correct Answer: (B) $c = 2$

Check conditions: $f(1) = 1 - 4 + 3 = 0$, $f(3) = 9 - 12 + 3 = 0$. Since $f(1) = f(3) = 0$, Rolle's theorem applies.

$f'(x) = 2x - 4 = 0 \Rightarrow x = 2$. Since $2 \in (1, 3)$, $c = 2$. ✓

Q17. $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$ by L'Hôpital or Taylor series equals:

- (A) 0 (B) 1 (C) $\frac{1}{2}$ (D) 2

✓ Answer & Explanation

Correct Answer: (C) $\frac{1}{2}$

Using the Taylor expansion: $e^x = 1 + x + \frac{x^2}{2!} + O(x^3)$.

$$\frac{e^x - 1 - x}{x^2} = \frac{\frac{x^2}{2} + O(x^3)}{x^2} = \frac{1}{2} + O(x) \xrightarrow{x \rightarrow 0} \frac{1}{2}$$

Alternative (L'Hôpital applied twice): $\lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}$.

6 Riemann Integration

Core Concepts

Riemann Sum

$$S(P, f) = \sum_{i=1}^n f(x_i^*) \Delta x_i \text{ over partition } P.$$

Riemann Integrable

f is Riemann integrable on $[a, b]$ iff $U(P, f) - L(P, f) < \varepsilon$ for some P .

FTC Part 1

If $F(x) = \int_a^x f(t) dt$, then $F'(x) = f(x)$ (for f continuous).

FTC Part 2

$$\int_a^b f(x) dx = F(b) - F(a) \text{ where } F' = f.$$

Improper Integrals

Type I: infinite limits; Type II: unbounded integrand.

Key Integration Results

$$\int_0^{\infty} e^{-x} dx = 1 \quad (23)$$

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \quad (\text{Gaussian}) \quad (24)$$

$$\int_0^1 x^{p-1} (1-x)^{q-1} dx = B(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)} \quad (\text{Beta function}) \quad (25)$$

$$\int_0^{\infty} x^{n-1} e^{-x} dx = \Gamma(n), \quad \Gamma(n) = (n-1)! \text{ for } n \in \mathbb{N} \quad (26)$$

6.1 MCQs: Integration

Q18. $\int_0^{\pi/2} \sin^2 x dx$ equals:

(A) 1 (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{4}$ (D) 0

Answer & Explanation

Correct Answer: (C) $\frac{\pi}{4}$

Using the identity $\sin^2 x = \frac{1 - \cos 2x}{2}$:

$$\int_0^{\pi/2} \sin^2 x \, dx = \int_0^{\pi/2} \frac{1 - \cos 2x}{2} \, dx = \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi/2} = \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{4}$$

Q19. The improper integral $\int_1^\infty \frac{1}{x^p} \, dx$ converges iff:

- (A) $p > 0$ (B) $p > 1$ (C) $p \geq 1$ (D) $0 < p < 1$

✓ Answer & Explanation

Correct Answer: (B) $p > 1$

$$\int_1^R \frac{dx}{x^p} = \begin{cases} \frac{R^{1-p} - 1}{1-p} & p \neq 1 \\ \ln R & p = 1 \end{cases}$$

As $R \rightarrow \infty$: converges iff $1 - p < 0$, i.e., $p > 1$. For $p \leq 1$ the integral diverges.

Q20. If f is continuous on $[a, b]$, the function $F(x) = \int_a^x f(t) \, dt$ satisfies:

- (A) $F'(x) = f(x)$ (B) $F'(x) = f(a)$ (C) $F'(x) = F(x)$ (D) F is not necessarily differentiable

✓ Answer & Explanation

Correct Answer: (A) $F'(x) = f(x)$

This is **Part 1 of the Fundamental Theorem of Calculus**. If f is continuous, then F is differentiable everywhere on (a, b) and $F'(x) = f(x)$.

7 Sequences and Series of Functions

📖 Pointwise vs Uniform Convergence

Pointwise $f_n \rightarrow f$ pointwise: for each fixed x , $|f_n(x) - f(x)| \rightarrow 0$.

Uniform $f_n \rightarrow f$ uniformly: $\sup_x |f_n(x) - f(x)| \rightarrow 0$ (one N works for *all* x).

Weierstrass M-test

If $|f_n(x)| \leq M_n$ and $\sum M_n < \infty$, then $\sum f_n$ converges uniformly.

Key Preservation

Uniform convergence preserves: continuity, integrability, and (with

extra condition) differentiability.

⚠ PSC Trap: Uniform vs Pointwise

$$f_n(x) = x^n \text{ on } [0, 1]: \text{ pointwise limit is } f(x) = \begin{cases} 0 & 0 \leq x < 1 \\ 1 & x = 1 \end{cases}.$$

The convergence is **NOT uniform** (limit is discontinuous despite each f_n being continuous). **On** $[0, r]$ **for** $r < 1$, **convergence IS uniform**.

7.1 MCQs: Uniform Convergence

Q21. The sequence $f_n(x) = \frac{x}{n}$ on $\mathbb{R}$ converges:

(A) Pointwise but not uniformly to $f \equiv 0$ (B) Uniformly to $f \equiv 0$ (C) Only for $x \in [-1, 1]$ (D) Does not converge

✓ Answer & Explanation

Correct Answer: (A) Pointwise but not uniformly

For each fixed x : $\frac{x}{n} \rightarrow 0$. So $f_n \rightarrow 0$ pointwise.

For uniform convergence: $\sup_{x \in \mathbb{R}} \left| \frac{x}{n} \right| = \infty$ for every n . Since this does not go to 0, convergence is **not uniform on** $\mathbb{R}$.

On any bounded interval $[-M, M]$: $\sup \left| \frac{x}{n} \right| = \frac{M}{n} \rightarrow 0$, so it is uniform on bounded sets.

Q22. The power series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converges:

(A) Only for $|x| < 1$ (B) For all $x \in \mathbb{R}$ (C) Only for $x \geq 0$ (D) For $|x| \leq e$

✓ Answer & Explanation

Correct Answer: (B) For all $x \in \mathbb{R}$

The radius of convergence: $R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \frac{1/n!}{1/(n+1)!} = \lim_{n \rightarrow \infty} (n+1) = \infty$.

Infinite radius of convergence means the series $\sum \frac{x^n}{n!} = e^x$ converges for **all** $x \in \mathbb{R}$.

8 Functions of Several Variables

Core Concepts

Partial Derivative

$f_x = \frac{\partial f}{\partial x}$: differentiate w.r.t. x treating y as constant.

Clairaut's Theorem

If f_{xy} and f_{yx} are continuous, then $f_{xy} = f_{yx}$.

Chain Rule

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$

Jacobian $J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}.$

Critical Points

$f_x = f_y = 0$; classify using $D = f_{xx}f_{yy} - f_{xy}^2$.

Euler's Theorem

If f is homogeneous of degree n : $xf_x + yf_y = nf$.

Second Derivative Test for $f(x)$

Let $D = f_{xx}f_{yy} - (f_{xy})^2$ at a critical point (a, b) :

$$D > 0, f_{xx} > 0 \Rightarrow \text{local minimum} \quad (27)$$

$$D > 0, f_{xx} < 0 \Rightarrow \text{local maximum} \quad (28)$$

$$D < 0 \Rightarrow \text{saddle point} \quad (29)$$

$$D = 0 \Rightarrow \text{test inconclusive} \quad (30)$$

8.1 MCQs: Several Variables

Q23. If $f(x, y) = x^3 + y^3 - 3xy$, the critical points satisfy:

(A) $x = 0, y = 0$ only (B) $x = 1, y = 1$ only (C) $(0, 0)$ and $(1, 1)$ (D) No critical points

Answer & Explanation

Correct Answer: (C) $(0, 0)$ and $(1, 1)$

$$f_x = 3x^2 - 3y = 0 \Rightarrow y = x^2$$

$$f_y = 3y^2 - 3x = 0 \Rightarrow x = y^2$$

Substituting: $x = (x^2)^2 = x^4 \Rightarrow x^4 - x = 0 \Rightarrow x(x^3 - 1) = 0 \Rightarrow x = 0$ or $x = 1$.

Critical points: $(0, 0)$ and $(1, 1)$.

Q24. If $f(x, y) = x^2y^3$, then by Euler's theorem $xf_x + yf_y$ equals:
 (A) $2f$ (B) $3f$ (C) $5f$ (D) $6f$

✓ **Answer & Explanation**

Correct Answer: (C) $5f$

$f(x, y) = x^2y^3$ is homogeneous of degree $n = 2 + 3 = 5$.

By Euler's theorem: $xf_x + yf_y = n \cdot f = 5f$.

Verification: $f_x = 2xy^3$, $f_y = 3x^2y^2$.

$xf_x + yf_y = 2x^2y^3 + 3x^2y^3 = 5x^2y^3 = 5f$. ✓

9 Additional High-Frequency PSC Topics

⚠ **Top 10 MCQ-Heavy Topics in Past AJKPSC Papers**

1. Standard limits: $\frac{\sin x}{x}$, $\left(1 + \frac{1}{n}\right)^n$, $\frac{e^x - 1}{x}$
2. p -series and comparison test
3. Continuity vs uniform continuity
4. Rolle's theorem and MVT (finding c)
5. Taylor/Maclaurin series of e^x , $\sin x$, $\cos x$, $\ln(1 + x)$
6. FTC Part 1 and Part 2
7. Convergence/divergence of improper integrals
8. Radius of convergence of power series
9. Euler's theorem for homogeneous functions
10. ε - δ definition of limit (mostly conceptual MCQs)

9.1 Mixed MCQs: PSC Exam Style

Q25. The radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n \cdot 2^n}$ is:

- (A) 1 (B) 2 (C) $\frac{1}{2}$ (D) ∞

✓ **Answer & Explanation**

Correct Answer: (B) 2

Using the ratio test: $R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \frac{(n+1) \cdot 2^{n+1}}{n \cdot 2^n} = \lim_{n \rightarrow \infty} 2 \cdot \frac{n+1}{n} = 2$.

Or use $R = \frac{1}{\limsup |a_n|^{1/n}} = \frac{1}{\lim (n \cdot 2^n)^{-1/n}} = \frac{1}{1/2} = 2$.

Q26. The function $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$ is:

(A) Differentiable at $x = 0$ (B) Discontinuous at $x = 0$ (C) Continuous at $x = 0$ (D) Neither continuous nor bounded near 0

✓ **Answer & Explanation**

Correct Answer: (C) Continuous at $x = 0$

Since $|x \sin(1/x)| \leq |x| \rightarrow 0$ as $x \rightarrow 0$, we have $\lim_{x \rightarrow 0} f(x) = 0 = f(0)$. So f is **continuous**.

However, f is **not differentiable** at 0: $\frac{f(x) - f(0)}{x} = \sin(1/x)$, which has no limit as $x \rightarrow 0$.

Q27. $\int_0^1 \frac{dx}{\sqrt{x}}$ is:

(A) Divergent (B) Equal to 1 (C) Equal to 2 (D) Equal to $\frac{1}{2}$

✓ **Answer & Explanation**

Correct Answer: (C) Equal to 2

This is a **Type II improper integral** (unbounded at $x = 0$):

$$\int_0^1 x^{-1/2} dx = \lim_{\varepsilon \rightarrow 0^+} \int_{\varepsilon}^1 x^{-1/2} dx = \lim_{\varepsilon \rightarrow 0^+} [2\sqrt{x}]_{\varepsilon}^1 = \lim_{\varepsilon \rightarrow 0^+} (2 - 2\sqrt{\varepsilon}) = 2$$

The integrand is unbounded but the integral **converges**.

Q28. The series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n}$ converges by:

(A) Ratio test (B) p -series test (C) Alternating series test (D) Comparison test

✓ Answer & Explanation

Correct Answer: (C) Alternating series test

This is the **alternating harmonic series** $1 - \frac{1}{2} + \frac{1}{3} - \dots = \ln 2$.

Conditions for Leibniz (alternating series) test: (i) $b_n = \frac{1}{n}$ is decreasing, (ii) $b_n \rightarrow 0$.

Both satisfied $\Rightarrow$ converges.

Note: it is **conditionally convergent** (not absolutely, since $\sum \frac{1}{n}$ diverges).

10 Master Summary Table

Topic	Key Result / Theorem	PSC Exam Tip
Sup / Inf	Every nonempty bounded set has sup and inf in $\mathbb{R}$	Sup may not be in the set
Sequences	Monotone bounded $\Rightarrow$ converges	Cauchy $\Leftrightarrow$ convergent in $\mathbb{R}$
p -Series	Converges iff $p > 1$	$\sum 1/n$ diverges!
Limits	$\frac{\sin x}{x} \rightarrow 1$, $\frac{e^x - 1}{x} \rightarrow 1$	Standard limits: memorise
Continuity	Cont. on $[a, b] \Rightarrow$ uniformly continuous	IVT, EVT on closed intervals
MVT	$f'(c) = \frac{f(b) - f(a)}{b - a}$	First check hypotheses
Taylor Series	$e^x, \sin x, \cos x, \ln(1 + x)$	Know first 4 terms
FTC	$\frac{d}{dx} \int_a^x f(t) dt = f(x)$	Most tested result
Improper Integrals	$\int_1^\infty x^{-p} dx$ converges iff $p > 1$	Type I vs Type II
Uniform Convergence	Weierstrass M-test	Uniform $\Rightarrow$ continuous limit
Homogeneous Functions	$xf_x + yf_y = nf$ (Euler)	Degree = sum of powers

★ Final Exam Strategy — Zamir Hussain

1. **Memorise** all standard limits and Taylor series — they appear in ≥ 4 MCQs.
2. **Identify** series type first (geometric, p -series, alternating) before applying a test.
3. **Always check** hypotheses of MVT and Rolle's theorem before applying.
4. For **improper integrals**: determine type (I or II) then evaluate with a limit.
5. In **uniform convergence**: compute $\sup_x |f_n(x) - f(x)|$ — if it $\rightarrow 0$, convergence is uniform.
6. **Homogeneous function** problems: find degree first, then Euler's theorem does the rest.
7. Practice **ε - δ proofs** — even conceptual understanding earns marks.

Exam Practice Resource

For further exam practice, the following resource provides a comprehensive collection of solved MCQs:

[Access the material here](#)

Best of Luck — PSC BPS-17 Mathematics Lecturer

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